

Notes for the talk

Lagrangian fibrations on Nikulin orbifolds

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These are notes for the talk and may contain inaccuracies.
The $\text{\LaTeX}$ source of these slides was prepared with the help of AI, starting from handwritten notes.

Lagrangian fibrations on Nikulin orbifolds

Theorem (Nanni 2025)

There are exactly 2 deformation families of Lagrangian fibrations on Nikulin-type orbifolds.

Corollary

The SYZ conjecture holds for Nikulin-type orbifolds.

Outline

- 1 IHS manifolds and the SYZ conjecture
- 2 Nikulin orbifolds and their Lagrangian fibrations
- 3 Deforming Lagrangian fibrations
- 4 Monodromy orbits and classification
- 5 References

K3 surfaces

Definition

A **K3 surface** is a compact complex Kähler surface S such that

- $\pi_1(S) = 0$;
- $H^0(S, \Omega_S^2) = \mathbb{C}\sigma$, with σ symplectic.

Example: $V(f_4) \subset \mathbb{P}^3$, a smooth quartic. **Fact:** $(H^2(S, \mathbb{Z}), \cap) \stackrel{\varphi}{\cong} \Lambda$ (marking).

Theorem (Torelli; Pyatetskii-Shapiro–Shafarevich 1971, Burns–Rapoport 1975)

There is a moduli space $\mathcal{M}_\Lambda$ of marked K3 surfaces, and the period map

$$\begin{array}{ccc} \mathcal{P}: & \mathcal{M}_\Lambda & \longrightarrow D \subset \mathbb{P}(\Lambda_{\mathbb{C}}) \\ & (S, \varphi) & \longmapsto \varphi[\mathbb{C}\sigma] \end{array}$$

induces an isomorphism on each connected component of the Hausdorff reduction of $\mathcal{M}_\Lambda$.

$$D = \{x : x^2 = 0, \ x\bar{x} > 0\}.$$

Irreducible holomorphic symplectic manifolds

Definition

An **irreducible holomorphic symplectic (IHS) manifold** is a compact complex Kähler manifold X such that

- $\pi_1(X) = 0$;
- $H^0(X, \Omega_X^2) = \mathbb{C}\sigma$, with σ symplectic.

Def/Prop (Beauville 1983, Fujiki 1987). The **Beauville–Bogomolov–Fujiki form** is the “unique” quadratic form q_X on $H^2(X, \mathbb{Z})$ such that there is $c_X \in \mathbb{R}_{>0}$ with

$$c_X q_X(\alpha)^n = \int_X \alpha^{2n} \quad \forall \alpha \in H^2(X, \mathbb{Z}), \quad 2n = \dim_{\mathbb{C}} X.$$

Theorem (Verbitsky 2013; Huybrechts 2012)

Fix a lattice Λ . The period map $\mathcal{P}: \mathcal{M}_\Lambda \rightarrow D \subset \mathbb{P}(\Lambda_{\mathbb{C}})$ of marked IHS manifolds of lattice Λ (defined as for K3 surfaces) induces an isomorphism on each connected component of the Hausdorff reduction of $\mathcal{M}_\Lambda$.

Fibrations of IHS manifolds

Theorem (Matsushita 1999)

A fibration $\psi: X \rightarrow B$ (surjective, connected fibres, $0 < \dim B < 2n$) satisfies:

- $\dim B = n$;
- $\rho(B) = 1$: $\text{Pic}(B)_{\mathbb{Q}} = \mathbb{Q}H$, H ample;
- the general fibre is a Lagrangian abelian variety.

$\ell = \psi^* c_1(H)$ is **algebraic, nef, isotropic**:

$$c_X q_X(\ell)^n = \int_X \psi^* c_1(H)^{2n} = \int_B H^{2n} = 0 \quad \Rightarrow \quad q_X(\ell) = 0.$$

Corollary

The very general IHS does **not** admit a fibration. Idea: $\sigma \in \bigcup_{\ell^2=0} \ell^\perp$, $\ell \in \Lambda$ (countable union).

The SYZ conjecture

Conjecture (SYZ; Verbitsky 2010)

If X admits an algebraic nef isotropic class ℓ , X admits a Lagrangian fibration with class a multiple of ℓ .

SYZ = Strominger–Yau–Zaslow (*Mirror symmetry is T-duality*, 1996)

Known: $K3^{[n]}$ [Bayer–Macrì 2014], Kum_n [Yoshioka 2016], OG6 [Mongardi–Rapagnetta 2021], OG10 [Mongardi–Onorati 2022].

Examples of IHS [Beauville 1983, Fujiki 1983]

$$\begin{array}{ccc} S \times S & & \text{too many 2-forms: } \lambda_1 \sigma \boxplus \lambda_2 \sigma \\ \downarrow / \mathfrak{S}_2 & & \downarrow \text{wavy} \\ S^{[2]} \xrightarrow{\text{Bl}} S^{(2)} & & \text{only } \lambda(\sigma \boxplus \sigma) \text{ descends} \end{array}$$

Symplectic orbifolds

Definition

An **orbifold** is a complex space locally isomorphic to quotients V/G , $V \subseteq \mathbb{C}^{2n}$ open, G a finite group.

Definition

An **irreducible symplectic orbifold** is a compact complex Kähler orbifold Y such that

- $\pi_1(Y_{\text{reg}}) = 0$;
- $H^0(Y, \Omega^2) = \mathbb{C}\sigma$, σ symplectic;
- $\text{codim Sing } Y > 2$.

Torelli theorem [Menet 2020]

Lagrangian fibrations [Schwald 2020]

Nikulin orbifolds

Let S be a K3 surface and ι a symplectic involution on S .

$$\begin{array}{ccccc}
 & & S^{[2]} & & \iota^{[2]} \text{ symplectic involution} \\
 & & \downarrow & & \\
 \text{Sing } Y: 28 \text{ points} & Y \xrightarrow{\text{Bl}_\Sigma} & S^{[2]}/\iota^{[2]} & \text{Sing}(S^{[2]}/\iota^{[2]}) : & \begin{cases} \Sigma \text{ K3 surface} \\ 28 = \binom{8}{2} \text{ points} \end{cases}
 \end{array}$$

8 fixed points of ι on S [Nikulin 1979]; general case [Mongardi 2012]

Definition

Such Y are called **Nikulin orbifolds**; their (locally trivial) deformations are of **Nikulin type**.

BBF lattice: $H^2(Y, \mathbb{Z}) \cong U(2)^3 \oplus E_8(-1) \oplus \langle -2 \rangle^2$, $c_Y = 6$ [Menet 2015].

Examples of fibrations

$\varepsilon: S \rightarrow \mathbb{P}^1$ elliptic, ι_+, ι_- symplectic involutions (e.g. on the Fermat quartic)

$$\varepsilon \circ \iota_+ = \varepsilon$$

$$\begin{array}{ccc} S^{[2]} & \longrightarrow & \mathbb{P}^2 \\ \downarrow & \nearrow & \\ Y_+ & \longrightarrow & S^{[2]}/\iota_+ \end{array}$$

fibre of type (1,2) [Barth 1987]; $\text{div } \ell = 1$

$$\varepsilon \circ \iota_- = \tau \circ \varepsilon$$

$$\begin{array}{ccc} S^{[2]} & \longrightarrow & \mathbb{P}^2 \\ \downarrow & & \downarrow \\ Y_- & \longrightarrow & S^{[2]}/\iota_- \longrightarrow \mathbb{P}^2/\tau \end{array}$$

fibre of type (1,1) [Wieneck 2016]; $\text{div } \ell = 2$

Different invariants $\Rightarrow$ not deformation equivalent.

Deforming: monodromy

A Lagrangian fibration $\psi: X \rightarrow B$ is the same as a semiample (isotropic) line bundle L on X .

Take a deformation (locally trivial) of X :

$$\begin{array}{ccc} X & \hookrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ 0 & \in & T \end{array}$$

Definition

A **deformation** of ψ is such a deformation together with a line bundle $\mathcal{L}$ on $\mathcal{X}$ with $\mathcal{L}|_X = L$, such that $L_t := \mathcal{L}|_{X_t}$ is semiample for all $t \in T$.

Introduce markings. Assume $\pi_1(T) = 0$ and fix a marking $\varphi: H^2(X, \mathbb{Z}) \rightarrow \Lambda$.

- For $t \in T$ choose a path γ_t from 0 to t : parallel transport $P_{\gamma_t}: H^2(X, \mathbb{Z}) \rightarrow H^2(X_t, \mathbb{Z})$, and set $\varphi_t := \varphi \circ P_{\gamma_t}^{-1}$.
- φ_t does not depend on γ_t : $\varphi_t^{-1} \circ \varphi'_t = P_{\gamma_t} P_{\gamma'_t}^{-1} = P_{\gamma_t(\gamma'_t)^{-1}} = \text{id}$, since $\gamma_t(\gamma'_t)^{-1} \in \pi_1(T) = 0$.

Deforming along paths

Remark. If $(\mathcal{X}, \mathcal{L})$ deforms (X, L) , then $\varphi_t(c_1(L_t)) = \ell \in \Lambda$
 $\forall t \in T \rightsquigarrow \mathcal{P}(T) \subseteq \ell^\perp$.

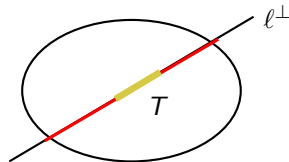
Theorem (Matsushita 2016;
Engel–Filipazzi–Greer–Mauri–Svaldi 2025)

A Lagrangian fibration (X, L) deforms over the locus where ℓ stays algebraic $(\mathcal{P}(T) \cap \ell^\perp)$, for T a small neighbourhood of 0.

What if $\pi_1(T) \neq 0$? Then the choice of γ_t matters! So fix a γ_t .

Theorem (smooth: Markman 2014, Matsushita 2017; singular: Onorati–Ríos Ortiz 2025)

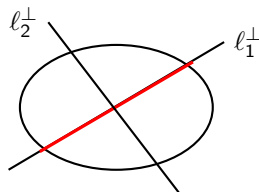
If ℓ stays algebraic and nef along γ_t , (X, L) deforms along γ_t . Moreover, if ℓ is algebraic and nef at the extremes, then there is a deformation of (X, L) containing X_t .



The monodromy group

How to reach other components?

We still have freedom on the marking.



Definition (Markman 2011)

The **monodromy group** $\text{Mon}^2(X)$ of X is the group of parallel transport operators P_γ , $\gamma \in \pi_1(T)$, for some deformation $\mathcal{X} \rightarrow T$.

Fact. If (X, φ_1) , (X, φ_2) are in the same connected component of $\mathcal{M}_\Lambda$, then

$$\varphi_2^{-1} \circ \varphi_1 \in \text{Mon}^2(X).$$

$\Rightarrow$ the class of a fibration can be moved anywhere in its Mon^2 -orbit.

How many orbits?

Isotropic classes up to monodromy: at most 3 orbits [Menet–Riess 2022]

Theorem (Brandhorst–Menet–Muller 2024; independently Nanni 2025)

Mon²(Y) is maximal: $\text{Mon}^2(Y) = O^+(H^2(Y, \mathbb{Z}))$.

Corollary

Exactly 2 orbits of primitive isotropic classes, distinguished by the divisibility (1 or 2).

Both are realised: Y_+ (div = 1) and Y_- (div = 2).

Theorem (Nanni 2025)

There are 2 deformation families of Lagrangian fibrations on Nikulin-type orbifolds.

Sketch of proof

- Take any (Y, L) . Take $(\hat{Y}, \hat{L})$ the one example with the same invariants (div of the primitive class).
- Fix a marking φ on Y , deform it to a marking $\hat{\varphi}$ on $\hat{Y}$.
- $\exists P_\gamma \in \text{Mon}^2(\hat{Y})$ such that $\hat{\varphi} \circ P_\gamma(\hat{L}) = \varphi(L)$.
- $\Rightarrow$ by [Onorati–Ríos Ortiz 2025], (Y, L) and $(\hat{Y}, \hat{L})$ are deformation equivalent as fibrations.

Corollary (of the proof)

SYZ: start from a nef isotropic algebraic class instead of L ; the same argument shows it is semiample.

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